Lab 10: Creating Alerts in Splunk

⚠️ **WARNING**: This lab requires a Splunk Enterprise trial or paid license. It will not work with the free license version.

Lab Overview

In this lab, you will learn to create intelligent alerts in Splunk that monitor your data in real-time and notify you when specific conditions occur. You'll build security alerts for failed login attempts, configure alert conditions and actions, and test the alerting system to ensure proper functionality.

Learning Objectives

By the end of this lab, you will be able to:

- Create real-time alerts based on search criteria
- Configure alert triggers and conditions
- Set up alert throttling to prevent alert storms
- Use the _audit index to monitor Splunk internal events
- Configure alert actions and severity levels
- Test alerts by triggering conditions
- View and manage triggered alerts
- Understand alert types (real-time vs scheduled)

Prerequisites

- Completed Labs 1-8
- Splunk Enterprise trial or paid license (NOT free license)
- Admin access to Splunk instance
- Understanding of Splunk search commands
- Familiarity with authentication concepts

> **Data window**: This lab watches the live `_audit` index, so it uses short, recent ranges (**Last 15 minutes**) rather than the window the uploaded course data covers. You generate the events yourself as you go.

Lab Environment Setup

Step 1: Verify License Type

Check your license type:

1. Navigate to **Settings** → **Licensing**
2. Ensure you have an **Enterprise** or **Trial** license
3. If you see **Free**, this lab cannot be completed

Step 2: Verify Admin Access

Ensure you have administrative privileges:

1. Check that you can access **Settings** menu
2. Verify you can see the _**audit** index
3. Confirm you can create alerts

Step 3: Enable Audit Logging

Ensure audit logging is enabled:

1. Navigate to **Settings** → **Server Settings** → **General Settings**
2. Verify **Enable Splunk Web SSL** is configured
3. Restart Splunk if needed

Key Concepts

What are Alerts?

Alerts are automated notifications triggered when specific conditions are met:

- Monitor data in real-time or on schedule
- Send notifications via email, webhook, or custom scripts
- Take automated actions like running scripts
- Integrate with incident management systems

Alert Types

Type	Description	Use Case
Real-time	Continuous monitoring	Security incidents, system failures
Scheduled	Run at specific intervals	Daily reports, batch processing

Alert Conditions

Condition	Description	Example
Number of Results	Count of matching events	Alert if > 5 failed logins
Custom	Complex logic with eval	Alert if error rate > 10%
Rolling Window	Time-based aggregation	Alert if 10 errors in 5 minutes

The _audit Index

Splunk's _**audit** index contains:

- User login/logout events
- Search activity
- Configuration changes
- System events

Exercise 1: Security Alert for Failed Logins

Scenario: Monitor failed login attempts on the Splunk search head. Alert when there's more than one failed admin login within one minute.

Task 1.1: Understand Audit Data

Search for recent login events:

```
index=_audit action="login attempt"
| head 10
| table _time, user, action, info, src_ip
```

Expected Fields:

- **action**: Type of activity (login attempt, logout, etc.)
- **info**: Result (failed, succeeded)
- **user**: Username attempting login
- **src_ip**: Source IP address

Task 1.2: Create Failed Login Search

Search for failed admin login attempts:

```
index=_audit action="login attempt" info=failed user=admin
```

Time Range: Set to **Last 15 minutes**

Expected Results: Events showing failed login attempts for admin user

Task 1.3: Generate Test Data

To test the alert, we need to generate failed login attempts:

1. **Log out** of Splunk using **username** → **Logout**
2. Try to log in with:
   - **Username**: admin
   - **Password**: WrongPassword
3. Repeat this process 2-3 times to generate test data
4. Finally, log in with correct credentials

✓ **Checkpoint**: Failed login events appear in the _audit index

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Exercise 2: Create Real-Time Alert

Task 2.1: Save Search as Alert

Using your failed login search:

```
index=_audit action="login attempt" info=failed user=admin
```

1. Click **Save As** → **Alert**
2. Configure basic settings:
   - **Title**: `Splunk Web Login Attempts`
   - **Description**: `Monitor failed admin login attempts`
   - **Permissions**: `Shared in App`

Task 2.2: Configure Alert Type

Set alert type and timing:

- **Alert type**: Real-time
- **Time Range**: Last 15 minutes (for testing)

Task 2.3: Configure Trigger Conditions

Set when the alert should fire:

- **Trigger alert when**: Number of Results
- **is greater than**: 0
- **in**: 1 minute

This means the alert triggers immediately when any failed admin login occurs.

Task 2.4: Configure Alert Behavior

Set how the alert should behave:

- **Trigger**: For each result
- **Throttle**: ✓ Enable throttling
- **Suppress results containing field value**: host
- **Suppress triggering for**: 60 seconds

Why Throttling? Prevents alert storms by suppressing duplicate alerts for 60 seconds.

Task 2.5: Configure Alert Actions

Add actions when alert triggers:

1. Click **Add Actions**
2. Select **Add to Triggered Alerts**
3. Configure:
    - **Severity**: High
    - **Add to**: Triggered Alerts (enabled by default)

Task 2.6: Save Alert

1. Click **Save**
2. Click **View Alert** to see the alert configuration

✓ **Checkpoint**: Alert created and ready for testing

Exercise 3: Test and Validate Alert

Task 3.1: Trigger Alert Conditions

Generate events that should trigger the alert:

1. **Log out** of Splunk
2. Attempt to log in with wrong password **3 times**:
    - Username: admin
    - Password: WrongPassword
3. Log in with correct credentials

Task 3.2: View Triggered Alerts

Check if alerts were triggered:

1. From the Splunk bar, click **Activity** → **Triggered Alerts**
2. Ensure **Search & Reporting** is selected for App
3. Look for "Splunk Web Login Attempts" alerts

Expected Results: One or more triggered alerts with High severity

Task 3.3: Examine Alert Details

Click **View results** on a triggered alert to see:

- Events that caused the alert
- Timestamp when alert fired
- Search results that matched conditions

✓ **Checkpoint**: Alert successfully triggered and visible in Activity menu

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Exercise 4: Advanced Alert Configuration

Task 4.1: Create Scheduled Alert

Create a summary alert that runs every 5 minutes:

```
index=_audit action="login attempt" info=failed
| stats count by user, src_ip
| where count > 2
```

Configure as:

- **Alert type**: Scheduled
- **Schedule**: Every 5 minutes
- **Trigger condition**: Number of Results > 0

Task 4.2: Email Alert Action

Add email notification:

1. Edit your original alert
2. Add Actions → **Send email**
3. Configure:
   - **To**: your-email@domain.com
   - **Subject**: Failed Login Alert - *name*
   - **Message**: Include search results

Task 4.3: Webhook Alert Action

For integration with external systems:

1. Add Actions → **Webhook**
2. Configure:
   - **URL**: http://your-webhook-url.com/alert
   - **HTTP Method**: POST
   - **Format**: JSON

Challenge Exercises

Challenge 1: Multi-Condition Alert

Create an alert for suspicious user activity:

```
index=_audit
| bucket _time span=5m
| stats dc(user) as unique_users, count by _time
| where unique_users > 10 OR count > 100
```

Alert when:

- More than 10 different users active in 5 minutes, OR
- More than 100 total activities in 5 minutes

Challenge 2: Correlation Alert

Monitor for brute force attacks:

```
index=_audit action="login attempt" info=failed
| bucket _time span=1m
| stats count by _time, src_ip
| where count >= 5
```

Alert for 5+ failed logins from same IP within 1 minute.

Challenge 3: Application Security Alert

Using your web application data:

```
index=main sourcetype=access_combined_wcookie status>=400
| bucket _time span=5m
| stats count by _time, clientip, status
| where count > 10
```

Alert for high error rates from specific IPs.

Alert Management

Alert States

State	Description	Action
Enabled	Alert is active and monitoring	Normal operation
Disabled	Alert exists but not running	Edit to re-enable
Skipped	Missed scheduled run	Check system resources

Alert History

View alert history:

1. Navigate to **Settings** → **Searches, reports, and alerts**
2. Find your alert and click **Edit**
3. Click **View triggered alert history**

Alert Performance

Monitor alert performance:

- **Search execution time**: Should complete quickly
- **Alert frequency**: Avoid too frequent alerts
- **Resource usage**: Monitor system impact

Best Practices

Alert Design Principles

1. **Specific Conditions**: Make alerts actionable, not noisy
2. **Appropriate Timing**: Balance between quick detection and false positives
3. **Clear Documentation**: Include context in alert descriptions
4. **Test Thoroughly**: Validate alerts work before deploying

Alert Naming Conventions

Use descriptive, consistent names:

- Environment: `PROD_Failed_Login_Alert`
- Severity: `CRITICAL_Database_Down`
- Category: `SECURITY_Suspicious_Activity`

Throttling Strategy

Alert Type	Throttling Period	Reason
Critical Security	5-15 minutes	Immediate but not overwhelming
Performance Issues	30-60 minutes	Avoid alert fatigue
Informational	1-24 hours	Summary notifications

Alert Actions

Action Type	Use Case	Configuration Tips
Email	Human notifications	Use clear subject lines
Webhook	System integration	Include context in payload
Script	Automated remediation	Test scripts thoroughly
Triggered Alerts	Splunk dashboard	Always include for tracking

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Troubleshooting Guide

Common Issues and Solutions

Issue	Symptom	Solution
Alert not triggering	No triggered alerts shown	Check search syntax and time range
Too many alerts	Alert storm/fatigue	Increase throttling period
Alert delays	Late notifications	Check system resources, simplify search
Missing audit data	Empty _audit index	Verify audit logging is enabled
Permission errors	Cannot create alerts	Check admin privileges

Debug Alert Searches

Test alert search manually:

```
| rest /services/saved/searches
| search title="Splunk Web Login Attempts"
| table title, search, cron_schedule, is_scheduled
```

Check alert status:

```
| rest /services/alerts/fired_alerts
| table title, trigger_time, severity
```

Performance Optimization

1. **Limit Time Range**: Use specific time windows
2. **Efficient Searches**: Avoid wildcards and regex when possible
3. **Index Selection**: Target specific indexes
4. **Field Optimization**: Extract only needed fields

Cleanup

Remove Test Alerts

After completing the lab:

1. **Delete Alerts**:
   - Navigate to Settings → Searches, reports, and alerts
   - Find "Splunk Web Login Attempts"
   - Click Delete
2. **Clear Triggered Alerts**:
   - Go to Activity → Triggered Alerts
   - Clear old test alerts

Key Takeaways

✅ **You've learned to:**

- Create real-time alerts for security monitoring
- Configure alert triggers and conditions
- Implement throttling to prevent alert storms
- Use the _audit index for system monitoring

- Test alerts by generating trigger conditions
- Manage triggered alerts through the Activity menu
- Understand different alert types and their use cases

Next Steps

- Proceed to Lab 11: Capstone — End-to-End Investigation, which combines the searching, statistics, lookups, dashboards and alerting from Labs 2 through 10 into a single investigation
- Explore advanced alert actions (scripts, webhooks)
- Learn about alert clustering and correlation
- Study integration with SOAR platforms
- Practice with custom alert conditions using eval

Quick Reference

Essential Alert SPL

```
# Basic failed login monitoring
index=_audit action="login attempt" info=failed user=admin

# Brute force detection
index=_audit action="login attempt" info=failed
| bucket _time span=1m
| stats count by _time, user, src_ip
| where count >= 5

# System error monitoring
index=_internal log_level=ERROR
| stats count by component
| where count > 10
```

Alert Configuration Template

```
Title: [SEVERITY]_[CATEGORY]_[DESCRIPTION]
Description: Clear explanation of what triggers alert
Search: Specific, efficient SPL query
Time Range: Appropriate for detection speed
Trigger: Based on business requirements
Throttling: Prevent alert fatigue
Actions: Appropriate notification methods
```

REST API Commands

```
# List all alerts
| rest /services/saved/searches
| search is_scheduled=1
| table title, search, cron_schedule

# View triggered alerts
| rest /services/alerts/fired_alerts
| table title, trigger_time, severity, sid

# Check alert status
| rest /services/saved/searches/[alert_name]
| table title, disabled, next_scheduled_time
```

Answer Key

🔑 **Instructor answer key**

Exercise 1

- Search: `index=_audit action="login attempt" info=failed user=admin`
- Expected events: Failed login attempts for admin user

Exercise 2

- Alert name: Splunk Web Login Attempts
- Alert type: Real-time
- Trigger condition: Number of Results > 0
- Throttling: 60 seconds on host field

Exercise 3

- Trigger method: 3 failed logins with wrong password
- Result: Alerts appear in Activity → Triggered Alerts
- Severity: High

Key Fields in _audit

- action: login attempt, logout, search, etc.
- info: failed, succeeded
- user: Username
- src_ip: Source IP address